



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CIRCUIT DEBUGGING ASSESSMENT

ECA0405 – ANALOG CIRCUITS | CO3 | ASSESSMENT – 2

Student Name	Vishal P	Register No.	192512340
Class / Section	ECA0405	Date	17/08/2026
Faculty Name	Dr. P Dass	Duration	90 Minutes
Assessment Type	Circuit Debugging	Maximum Marks	20

Assessment Objective:

To identify, analyze, and correct errors in an analog circuit, justify the debugging approach, verify the corrected circuit operation, and relate the result to CO3

Question:

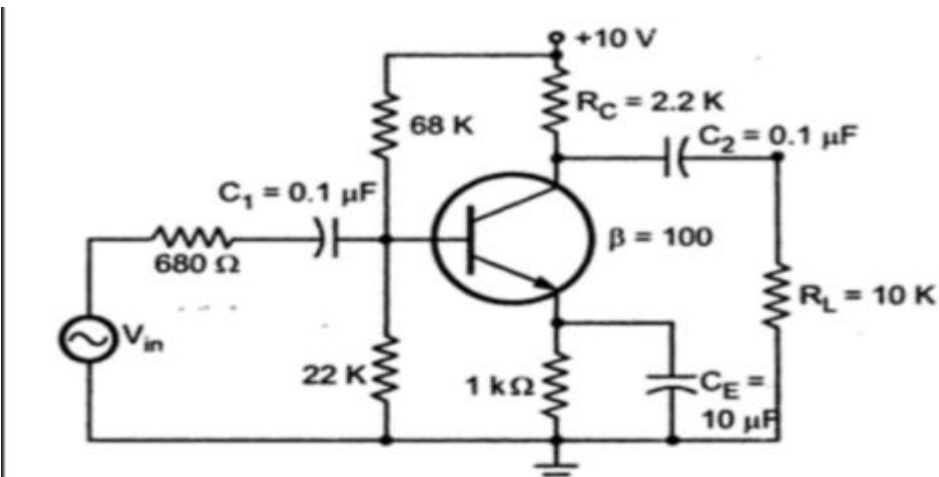
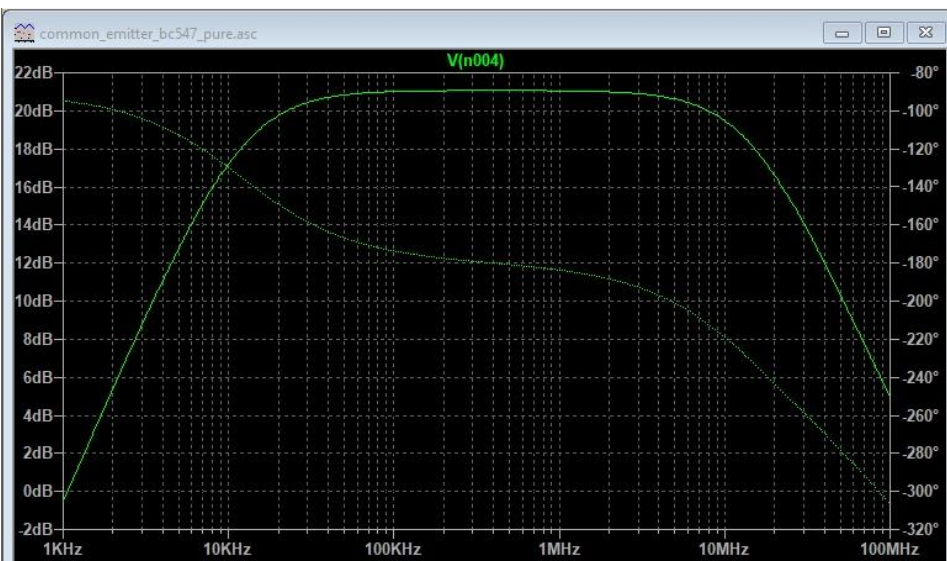
CO3 : Examine the frequency response characteristics of BJT and FET amplifiers at low and high frequencies to determine bandwidth, gain variation, and overall amplifier performance

CO3 AT2 Circuit Debugging:

Description:

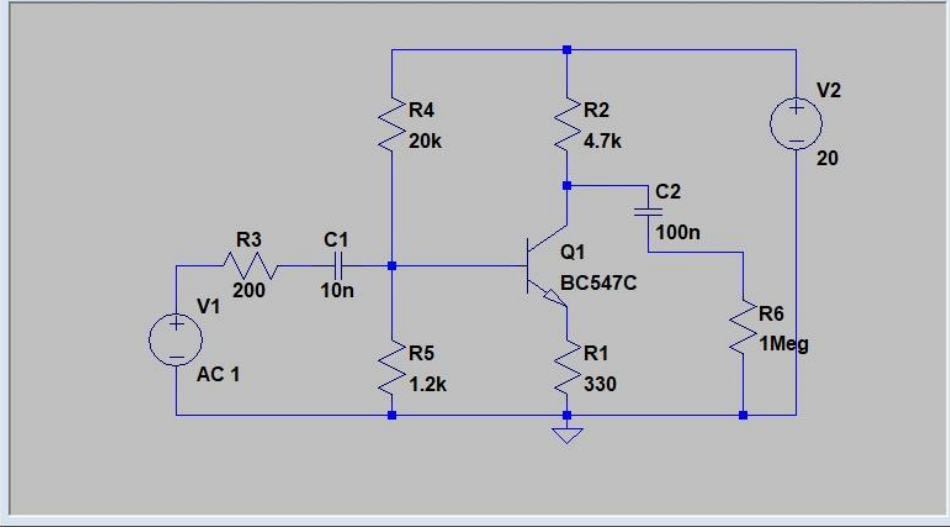
- Each student will be assigned one Circuit debugging problem 20 marks inside class. Analyze and debug one faulty analog circuit.
- Simulate the corrected circuit using LTspice and verify results.
- Document the debugging process, observations, and conclusions.
- Upload LTspice files, simulation results, and report to GitHub.
- Evaluation is based on fault identification, debugging approach, simulation accuracy, documentation, and GitHub submission.

Circuit Debugging Record

Item	Student Entry
1. Given Circuit / Circuit Diagram	Paste or draw the assigned analog circuit here. 
2. Expected Circuit Operation	
3. Observed Fault / Symptom	Lower cut-off frequency is observed as 420 Hz instead of the expected 250 Hz. This indicates that the amplifier's low-frequency response is degraded, causing the lower cut-off frequency to shift to a higher value.
4. Suspected Component / Connection	Emitter bypass capacitor CE = 10 uF is suspected to be faulty or having reduced effective capacitance. It may not be providing sufficient AC



	bypassing at low frequencies, causing the lower cut-off frequency to increase from 250 Hz to 420 Hz .
5. Debugging Analysis / Calculation	For each capacitor, the low-frequency cutoff is approximately: $f_L = \frac{1}{2\pi RC_1}$ For the emitter bypass capacitor: $R_{E(eq)} \approx R_E \parallel \frac{R_B}{\beta + 1} + r_{\pi}$ Using the circuit values, the emitter bypass capacitor has a significant effect on the low-frequency response. The observed value is: $f_L(\text{observed}) = 420\text{Hz}$ while the expected value is: $f_L(\text{expected}) = 250\text{Hz}$ Since a smaller effective capacitance increases f_L, the $10\mu\text{F}$ emitter bypass capacitor is suspected. LTspice verification: Keep $C_1 = 0.1\mu\text{F}$ and $C_2 = 0.1\mu\text{F}$ unchanged and increase C_E. The lower cutoff frequency shifts downward toward 250 Hz, confirming C_E as the likely faulty component.
6. Error Identified	The emitter bypass capacitor $C_E = 10\mu\text{F}$ has insufficient effective capacitance, resulting in inadequate AC bypassing of the emitter resistor at low frequencies. This shifts the lower cut-off frequency upward from the expected 250 Hz to 420 Hz .
7. Correction Made	The emitter bypass capacitor C_E was replaced/increased from $10\mu\text{F}$ to a higher suitable value . The circuit was re-simulated in LTspice, and the lower cut-off frequency shifted downward from 420 Hz toward the expected 250 Hz , confirming the correction.

8. Corrected Circuit Diagram	
9. Verification / Simulation Output	The corrected circuit was simulated using LTspice AC Sweep analysis. The simulation showed that the lower cut-off frequency shifted from approximately 420 Hz to 250 Hz after increasing the emitter bypass capacitor CE . Thus, the LTspice simulation confirms that the correction successfully restored the expected low-frequency response.
10. Final Result & Technical Justification	The faulty emitter bypass capacitor CE was identified and corrected. After increasing the capacitance and re-simulating the circuit in LTspice, the lower cut-off frequency decreased from 420 Hz toward the expected 250 Hz. This confirms that insufficient emitter bypassing was responsible for the increased lower cut-off frequency.



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Marks Distribution

Assessment Criterion	Marks	Marks Awarded	Faculty Remarks
Problem identification & circuit understanding	4		
Fault analysis & technical calculations	5		
Correct identification of error	3		
Effective circuit correction	3		
Verification / output validation	3		
Technical explanation & documentation	2		
Total	20		